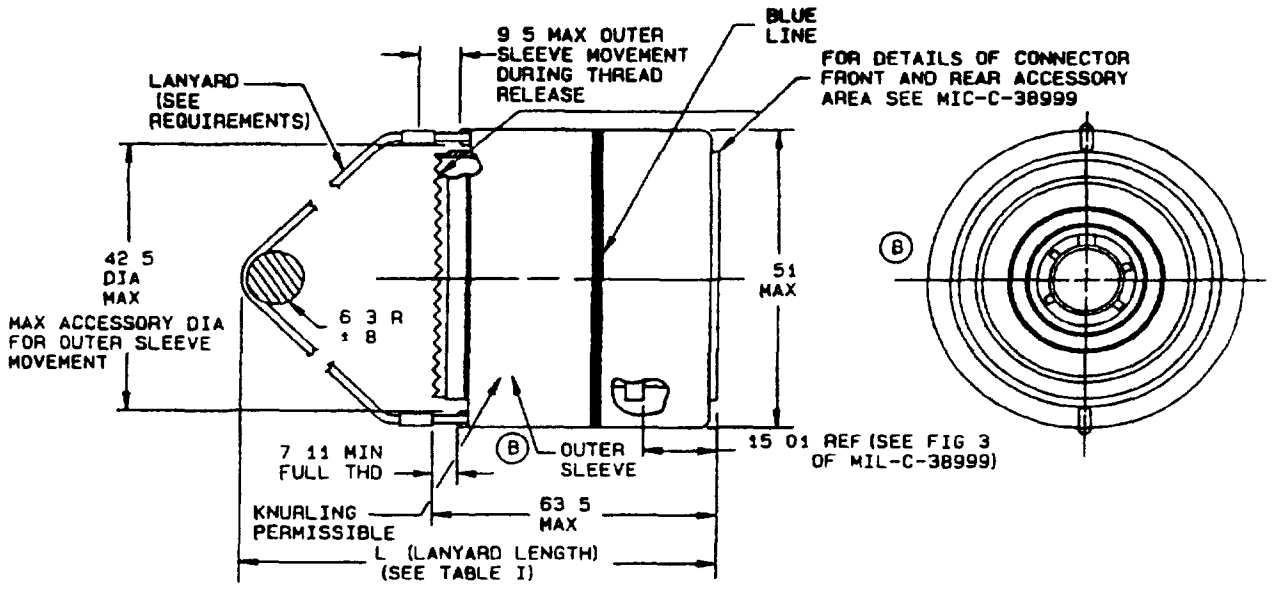
METRIC
MIL-C-38999/31B
10 June 1992
SUPERSEDING
DOD-C-38999/31A
13 NOVEMBER 1987

MILITARY SPECIFICATION SHEET

CONNECTORS, ELECTRICAL, CIRCULAR, PLUG, LANYARD RELEASE, FAIL-SAFE REMOVABLE CRIMP CONTACTS, PINS, SERIES III, SHELL SIZE 25, METRIC

This specification is approved for use by all Departments and Agencies of the Department of Defense

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation MIL-C-38999.



mm	Inch
6.3	0 250
7 11	0.280
9.5	0.370
15.01	0.591
42.5	1 67
51 0	2 00
63 5	2 50

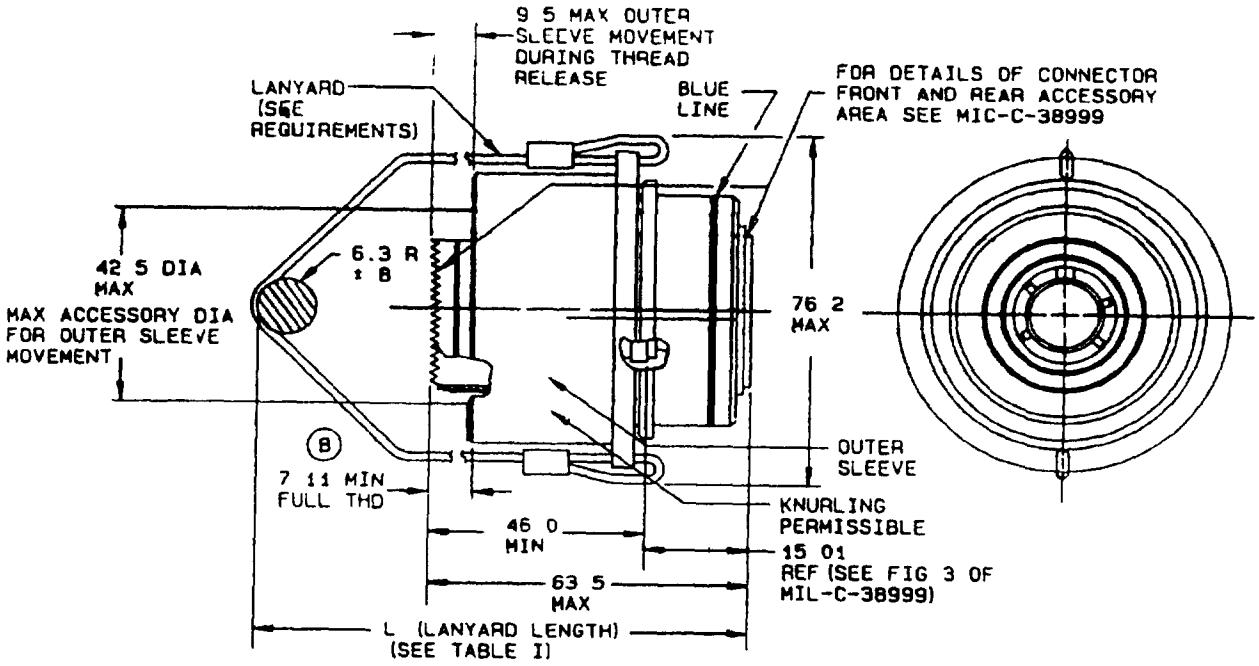
NOTES:
1. Dimensions are in millimeters
2. Inch equivalents are given for general information only.
3. Polarizing position N is shown.
4. EMI grounding feature required on this connector.
5. The Government may stock, store, and issue this configuration.

FIGURE 1. Plug, class W, type 1

(B) denotes changes

DISTRIBUTION STATEMENT A Approved for public release, distribution is unlimited

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mm	Inch
0 8	0 03
6.3	0 25
7 11	0.280
9 5	0.37
15 01	0 591
42 5	1 67
46.0	1.81
63 5	2.50
76.2	3.00

NOTES.
1. Dimensions are in millimeters
2. Inch equivalents are given for general information only
3. Polarizing position N is shown.
4. EMI grounding feature required on this connector
(8) 5 Inactive for new design, the Government will not stock, store, and issue this configuration

FIGURE 2 Plug, class W, type 2

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REQUIREMENTS

Dimensions and configurations: See figure 1 Interface dimensions shall conform to MIL-C-38999

This connector mates with and is designed for use with receptacle connectors MIL-C-38999/20 and /24

Connector shall accept MIL-C-85049 self-locking accessories

B For insert arrangements See MIL-STD-1560

Lanyard

a 1 57 mm minimum diameter, seven strands of stainless steel capable of withstanding 890 newtons pull test after assembly with the connector

b. Coupling design optional. Cable shall be covered with a suitable protective sleeving to preclude possible chaffing of wires.

Connector shall disengage from any coupling condition including partially mated.

Connector design shall incorporate a swivel action for the lanyard to prevent twisting of the cable

Spare contacts: The twin-axial and coaxial contacts are exempted from the spare contact requirements of MIL-C-38999

QUALIFICATION

Connectors shall meet the qualification requirements of MIL-C-38999 with the following exceptions and additions

Table XI, qualification and group and group C periodic tests (see MIL-C-38999)

Group 11 Replace coupling torque requirement with pull-separation force

Durability Wired connectors shall meet the durability requirements of MIL-C-38999, with the following exception

The total number of cycles of mating and unmating shall be 500, in the following sequence 200 cycles of normal mating and unmating, 50 cycles of normal mating with pull-separation unmating, 200 cycles of normal mating and unmating, and 50 cycles of normal mating with pull-separation unmating The lanyard release velocity during the pull-separation unmating cycles shall be 9.15 m/s

Pull-separation force In addition to mating and unmating by normal coupling ring rotation, the connector shall be capable of lanyard-pull separation at any angle within 15° of normal axis Each connector shall have one straight pull and one pull at 15° from straight, with a pull rate not exceeding 13 cm/s The test will be at -65°C (tolerance of +0°C, -5°C) at ambient, and at the maximum temperature of the specified class The test will be conducted within 3 minutes after removal from the temperature chamber without forced heating or cooling Maximum separation forces shall be 400 newtons for a straight pull and 445 newtons for a 15° pull.

Fail-safe disengagement: Connectors shall be partially mated with the plug coupling ring rotated 50 percent of full coupling Pull-separation shall be accomplished within the limits as specified above

Vibration: Wired mated connectors shall meet the vibration requirements of MIL-C-38999 with the following exceptions

a. Sine vibration: Connectors shall be subjected to the test specified in method 204, test condition G, of MIL-STD-202

b. Random vibration Connectors shall be subjected to the test specified in method 2005 of MIL-STD-1344, test condition VI, letter J, ambient temperature Duration shall be 8 hours in the longitudinal direction and in a perpendicular direction, for a total of 16 hours

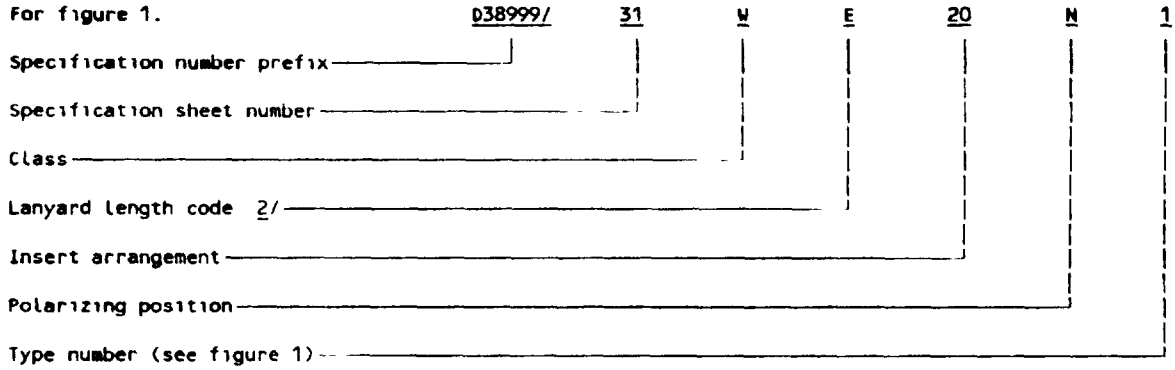
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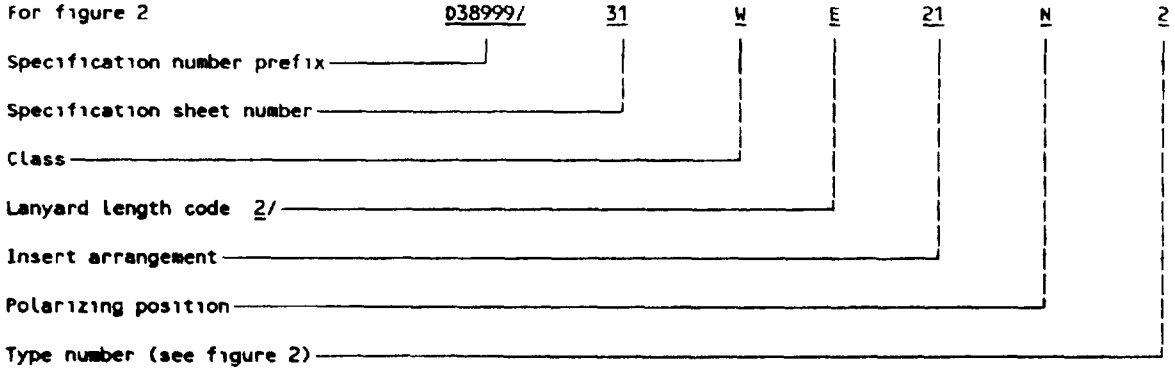
c The QPL evaluating activity will define the accessory load and cable to be used in the random and sine vibration tests

Ice resistance The mated, wired connectors with accessories attached shall be placed in a chamber and the temperature reduced and stabilized such that the item is maintained at -18°C (tolerance of +0°C, -5°C) for 1 hour. After stabilization of the chamfer temperature, the test item shall be spray with water precooled to 2°C (tolerance of +5°C, -0°C), for a period of 5 minutes The test item shall be located a maximum of 305 mm (12 inches) from the spray nozzle The entire test item shall be exposed to the spray After completion of water spray, the test item shall remain in the chamber -18°C (tolerance of +0°C, -5°C) for an additional 30 minutes Upon completion of the 30 minute cold soak period, the test item shall removed from the chamber and immediately (within 2 minutes) subjected to uncoupling by use of the lanyard mechanism The force required to separate the connector shall not exceed 400 newtons straight pull and 445 newtons 15° pull by more than 50 percent

Dust (fine sand) Upon completion of the dust test, the test items shall be removed from the chamber and immediately subjected to uncoupling by use of the lanyard mechanism The force required to separate the connectors shall not exceed the required values specified for straight pull separation by more than 25 percent

Part or Identifying Number (PIN) example·





NOTE The term Part or Identify Number (PIN) is equivalent to the term (part number, identification number, and type designator) which was previously used in this specification

1/ The Government may stock, store, and issue type 1 only (see figure 1)
2/ See table I

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Ⓑ TABLE I Lanyard length code

Code	L ± 6
E	153 (6 0)
F	166 (6 5)
G	178 (7 0)
H	191 (7 5)
I	203 (8 0)
J	216 (8 5)
K	229 (9 0)
L	242 (9 5)

CONCLUDING MATERIAL

Custodians
Army - CR
Navy - AS
Air Force - 85

Review activities
Ⓑ Army - MI
Navy - EC, MC, OS
Air Force - 11, 15, 99
DLA - ES

User activity
Air Force - 80

Preparing activity
Air Force - 85

Agent
DLA - ES

(Project 5935-3822)

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